

INTRODUCTION

This document presents the iterative development of the Amazon Products Review Analyser web application, built using the Streamlit framework and deployed on Hugging Face Spaces. The application integrates two independent machine learning models, a DistilBERT-based sentiment classifier and a Logistic Regression suspicious review detector into a single user facing interface designed to make model predictions transparent, explainable, and accessible to a nontechnical audience.

The development process was structured around a series of iterative stages, each representing a meaningful improvement to the application's functionality, user experience, and interpretability. Rather than building the final application in a single pass, the team adopted an agile approach, deploying working versions at each stage, gathering feedback, identifying limitations, and implementing targeted improvements.

The progression from Stage 1 to Stage 4 reflects the team's growing understanding of both the technical constraints of deploying machine learning models in real time and the practical requirements of building an interface that communicates model behaviour clearly. Key challenges addressed across the stages include the training-serving skew problem associated with the review frequency feature, the interpretability of model predictions through SHAP analysis, and the user experience design decisions required to present complex outputs in a digestible format.

STAGE 1: Initial Deployment

Sentiment Analysis only

Your Review

Type your review here...

Classify Review

Your Review

As always with Grisham a nice easy read.
Not quite top tier Grisham, but still enjoyable.
Nevertheless I would still recommend.

Classify Review

Sentiment: Positive

Confidence Score
0.6648

Higher absolute value = more confident. Positive score → Positive sentiment.

Your Review

Worked like a dream for 7 months and then stopped connecting and is unusable. I'd expect to use something like this for years not months. Such a waste and no way of recycling the materials properly.

Classify Review

Sentiment: Negative

Confidence Score
-0.9653

Higher absolute value = more confident. Positive score → Positive sentiment.

Overview

Stage 1 represented the very first deployable version of the web application. The primary objective at this stage was not to deliver a complete product but to establish a working local deployment pipeline and validate that the application ran correctly before any hosting decisions were made. The decision was made to deploy the sentiment model alone, while still fine tuning the suspicious review model, to confirm the infrastructure was stable before integrating additional complexity.

This approach is consistent with industry best practice for iterative software development, shipping a minimal working version first, verifying that it functions correctly in a local environment, and then building on that foundation. Deploying both models simultaneously in the first iteration would have made it significantly harder to isolate and debug any infrastructure issues that arose during deployment.

The Stage 1 interface was intentionally minimal. Users could enter a review in a text area and submit it for sentiment analysis. The output was a basic confidence score indicating the model's confidence in its prediction, alongside a binary Positive or Negative label. No additional visualisations, explanations, or structured feature outputs were included at this stage.

The simplicity of the Stage 1 interface was deliberate. At this stage, the application ran locally via Streamlit on localhost, with cloud hosting not yet decided upon. The team later evaluated cloud deployment options before settling on Hugging Face Spaces as the hosting platform in subsequent stages. The immediate focus was on confirming that the DistilBERT model loaded correctly, that inference ran within acceptable time limits on CPU hardware, and that the Streamlit application responded correctly to user input. These were non-trivial technical challenges given that transformer-based models such as DistilBERT are computationally demanding and required careful configuration to run efficiently without GPU support.

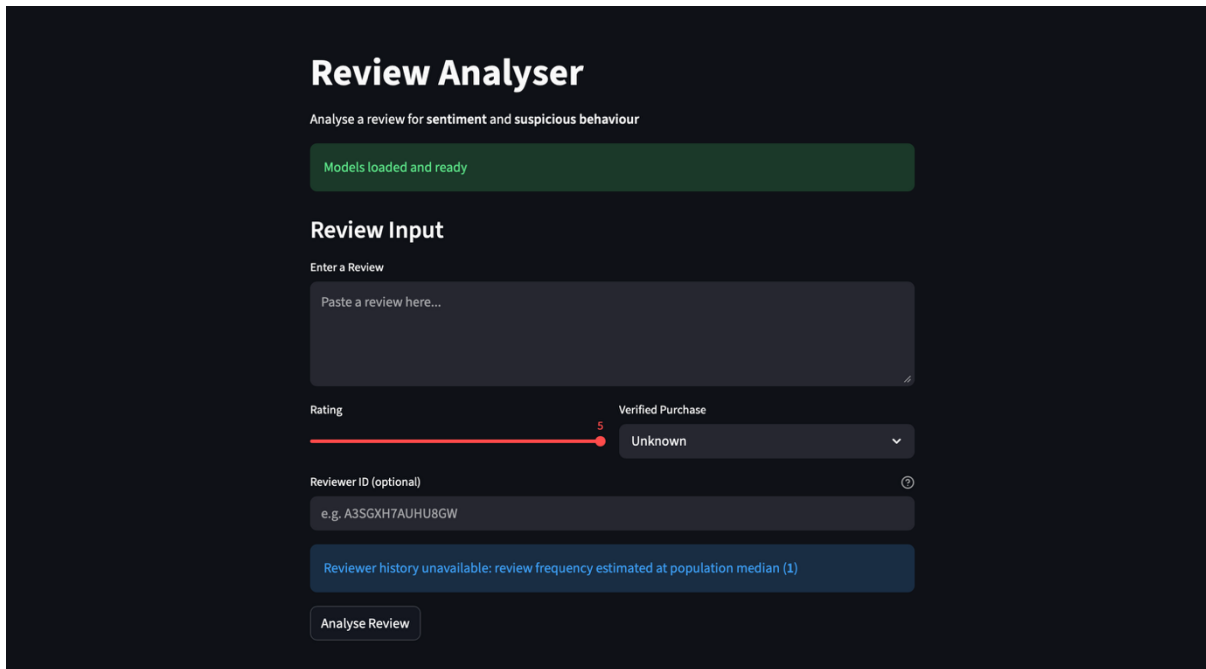
Key Feature at this stage

- Text area for review input
- Sentiment prediction label: Positive or Negative
- Confidence score indicating model confidence
- DistilBERT model loaded and served locally via Streamlit on localhost
- Basic Streamlit layout with no sidebar or additional UI elements

Limitations Identified

Stage 1 served its purpose as a proof of concept but had clear limitations that informed the next iteration. The suspicious review detection model was absent, meaning the application only addressed half of the project's core objective. The output provided no explanation of what drove the sentiment prediction, offering the user only a label and a number with no context. There were no visual elements, no structured metadata inputs such as rating or verified purchase status, and no reviewer frequency resolution logic. These gaps were systematically addressed in Stage 2.

STAGE 2: Both Models Integrated



Review Analyser

Analyse a review for sentiment and suspicious behaviour

Models loaded and ready

Review Input

Enter a Review

Paste a review here...

Rating 5 Verified Purchase Unknown

Reviewer ID (optional) ⓘ

e.g. A3SGXH7AUHU8GW

Reviewer history unavailable: review frequency estimated at population median (1)

Analyse Review

Overview

Stage 2 introduced the full dual model architecture, integrating the Logistic Regression suspicious review detector alongside the existing DistilBERT sentiment classifier. This stage marked the transition from a single purpose tool into a complete review analysis application capable of delivering both types of prediction from a single user submission. This stage also marked the transition from local development to cloud deployment, with the application moved from localhost to Hugging Face Spaces, making it publicly accessible for the first time.

The interface was extended to capture the additional inputs required by the suspicious review model, and the reviewer frequency resolution logic was implemented to address the training serving skew problem identified during model development.

Interface Design

The Stage 2 interface presented a clean, structured layout. The application's colour theme, Light or dark, is automatically determined by the user's device system preference, as Streamlit inherits the operating system display settings rather than applying a fixed theme.

A key design decision at this stage was the introduction of the reviewer frequency transparency note. When no Reviewer ID was provided, the application displayed an information banner informing the user that the review frequency feature, the most important feature in the suspicious review model based

on feature ablation, would be estimated using the population median from the training data. This was a principled and honest response to an inherent limitation: the model was trained on reviewer history that is unavailable for new reviews at inference time. Rather than silently imputing a value, the application made this substitution explicit and transparent.

Key Features at This Stage

- Both sentiment and suspicious review models integrated into a single interface
- Rating slider, verified purchase dropdown and optional Reviewer ID field added
- Reviewer frequency lookup table implemented, 630,543 reviewers from training data
- Population median fallback with transparent information notes when reviewer is unknown
- Application deployed to Hugging Face Spaces, publicly accessible for the first time
- Model loading confirmation displayed on startup
- Single Analyse Review button triggering both model predictions simultaneously

Limitations Identified

While Stage 2 delivered the complete dual model pipeline, the output remained limited in terms of interpretability and user experience. Users received prediction labels with no confidence scores, no visual representations, and no explanation of what features or words drove each prediction. The interface also lacked a sidebar for contextual information, a word count indicator, a clear button, and any structured breakdown of the model's reasoning. These limitations were addressed comprehensively in Stage 3.

STAGE 3: Explainability Result and Presentation

Review Analyser

Analyse a review for sentiment and suspicious behaviour

Models loaded and ready

Review Input

Enter a Review

Paste a review here...

Rating

5

Verified Purchase

Unknown

Reviewer ID (optional)

e.g. A3SGXH7AUHU8GW

Analyse Review

Review Analyser

Analyse a review for sentiment and suspicious behaviour

Models loaded and ready

Review Input

Enter a Review

physique.

Going up gentles slopes requires more leg power but I will sometimes just get off and push the bike up manually to reduce strain on the battery.

Balance seems to be good but beware as the small wheels can cause steering to tricky at times

Rating

5

Verified Purchase

Yes

Reviewer ID (optional)

e.g. A3SGXH7AUHU8GW

Reviewer ID (optional)

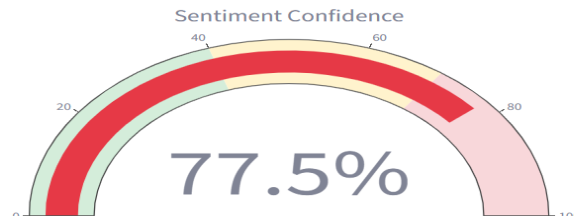
e.g. A3SGXH7AUHU8GW

Reviewer history unavailable: review frequency estimated at population median (1)

Analyse Review

Sentiment Analysis

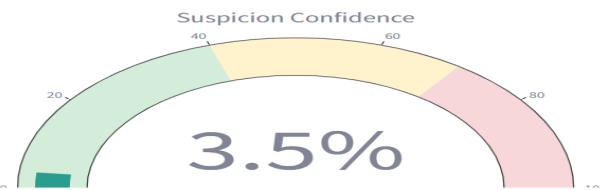
Sentiment: **Positive**



The model is 77.48% confident that this review sentiment is **likely Positive** as a result of the following words:

Suspicious Review Detection

Not Suspicious



The model is 3.5% confident that this review is **likely not suspicious** as a result of the following features:

Verified purchase: this review comes from a confirmed buyer, which reduces the likelihood of suspicious behaviour.

High extreme rating (5 stars): a 5 star rating was given. While extreme ratings are common in fake reviews, they are equally common among genuine enthusiastic buyers. On its own this is a weak signal, and the model treats it as a mild indicator rather than a strong flag.

Sufficient review length: the review contains enough words to analyse. Longer reviews generally provide more genuine detail and are less commonly associated with suspicious behaviour.

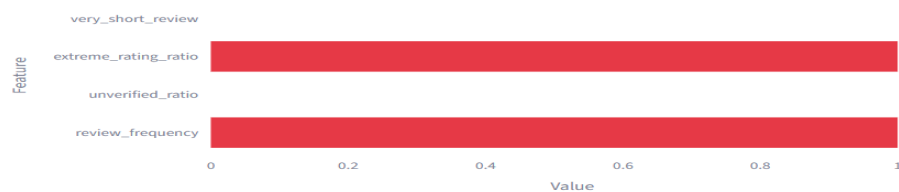
Sufficient review length: the review contains enough words to analyse. Longer reviews generally provide more genuine detail and are less commonly associated with suspicious behaviour.

Low reviewer volume (frequency: 1): this reviewer has posted very few reviews in total. Low volume reviewers are less likely to be engaged in coordinated or incentivised reviewing campaigns, which typically involve posting many reviews across multiple products.

Structured Feature Values for This Review

This chart shows the actual value each structured feature had for this specific review. All features are on a 0 to 1 scale. A value of 1 means the flag was fully triggered, 0 means it was not triggered, and 0.5 represents an unknown or neutral state. These values were passed directly into the model alongside the review text.

Structured Feature Values for This Review



Feature Impact (SHAP)

Feature Impact (SHAP)



What each feature contributed to this prediction

- **review_frequency** (+0.0000): slightly pushed **away** from suspicious. Higher reviewer volume increases suspicion as it mirrors the high frequency posting patterns seen in coordinated fake review campaigns.
- **unverified_ratio** (+0.0000): slightly pushed **away** from suspicious. Unverified purchases are more commonly linked to suspicious reviewing behaviour where the reviewer may not have actually bought the product.
- **extreme_rating_ratio** (+0.0000): slightly pushed **away** from suspicious. Extreme ratings (1 or 5 stars) are a weak signal, they appear in both genuine and suspicious reviews, so the model gives this moderate weight.
- **very_short_review** (-0.0000): slightly pushed **away** from suspicious. Very short reviews lack the detail

Overview

Stage 3 represented a significant step forward in the application's maturity. The primary focus of this iteration was explainability, ensuring that the predictions produced by both models were communicated clearly and meaningfully to the user. Rather than simply returning a label, the application now provided confidence scores, visual gauge charts, structured feature explanations, SHAP-based feature impact analysis, and plain-English interpretations of what drove each prediction.

Sentiment Analysis Output

The sentiment analysis section was significantly enhanced in Stage 3. In addition to the binary Positive or Negative label, the application now displayed a gauge chart showing the model's confidence percentage, providing an intuitive visual representation of how certain the model was in its prediction. The gauge used colour coding, teal for low confidence and red for high confidence, to make the confidence level immediately scannable without requiring the user to interpret a raw number.

Beneath the gauge, a plain-English interpretation sentence communicated the model's confidence in natural language, for example, stating that the model was 77.48% confident that the review sentiment was likely Positive because of specific words. This framing was chosen deliberately to bridge the gap between statistical output and human-readable explanation, making the result accessible to users without a machine learning background.

Suspicious Review Detection Output

The suspicious review detection section was similarly enhanced. A confidence gauge displayed the suspicion probability, and a structured plain-English explanation described the contribution of each of the four engineered features, verified purchase status, extreme rating, review length, and reviewer volume. Each feature was explained in context, helping the user understand not just what the model predicted but why those specific features influenced the outcome.

A structured feature values chart provided a visual breakdown of the actual numeric value each feature held for the review being analysed. All features were mapped to a 0 to 1 scale, where 1 indicated the flag was fully triggered and 0 indicated it was not. This chart allowed users to see immediately which features were active for any given review, grounding the model's explanation in concrete measurable values.

SHAP Feature Impact Analysis

One of the most significant additions in Stage 3 was the integration of SHAP (SHapley Additive exPlanations) analysis for the suspicious review model. A horizontal bar chart displayed the SHAP impact of each structured feature, colour coded in red for features that pushed the prediction toward suspicious and teal for features that pushed it away. The length of each bar indicated the strength of the feature's influence on that specific prediction.

Below the SHAP chart, a plain-English interpretation was provided for each feature, explaining the direction and magnitude of its contribution in accessible language. This combination of visual and textual explanation was designed to make the model's decision-making process as transparent as possible, supporting the project's broader objective of building a trustworthy and interpretable review analysis tool.

Key Features Added at This Stage

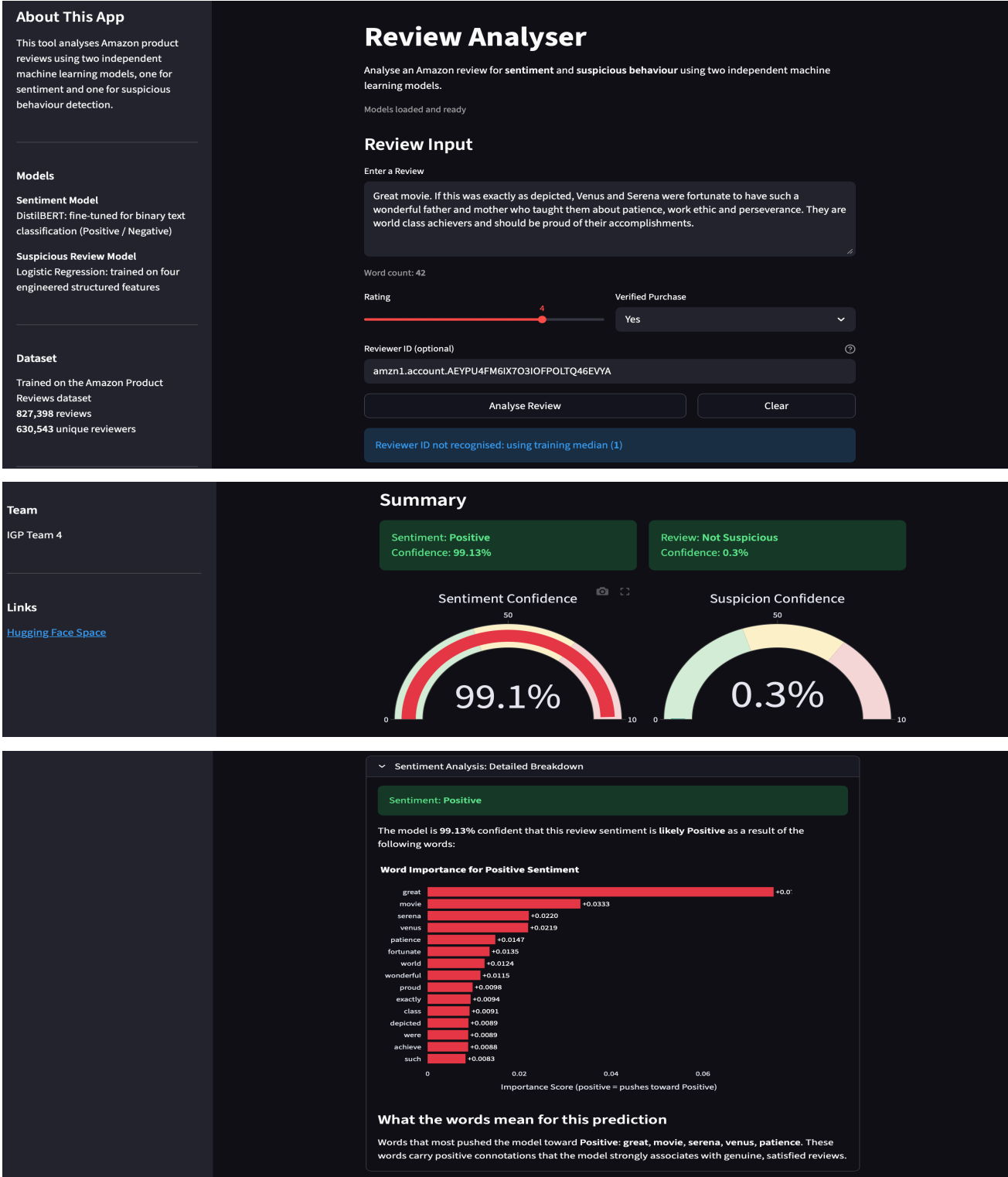
- Confidence gauge charts for both sentiment and suspicious review models
- Plain-English interpretation sentence for each model prediction
- Structured feature breakdown chart showing actual feature values on a 0 to 1 scale
- SHAP horizontal bar chart showing feature impact direction and magnitude
- Plain-English SHAP interpretation for each of the four structured features
- Reviewer frequency transparency note displayed after analysis is triggered

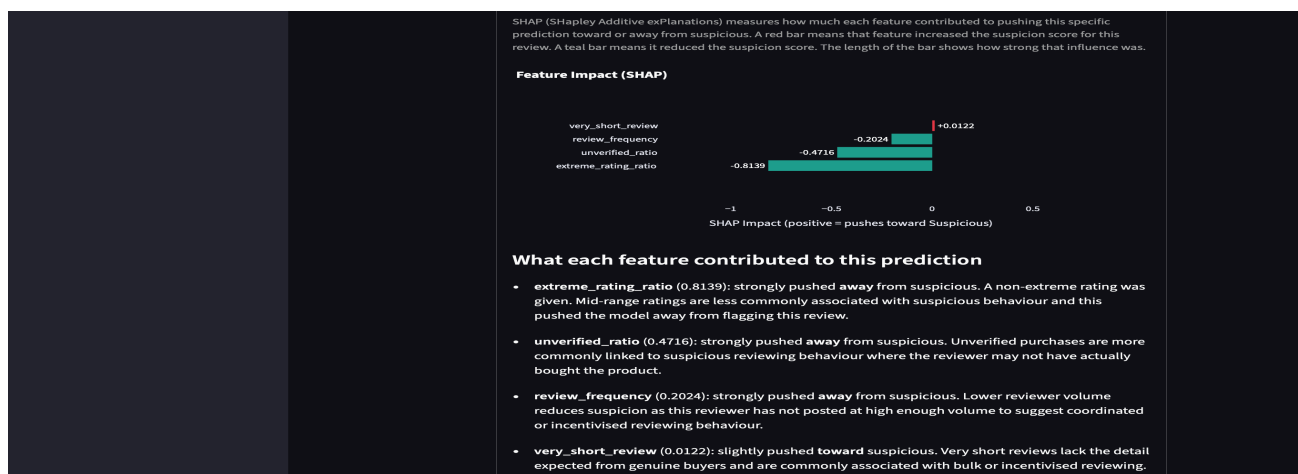
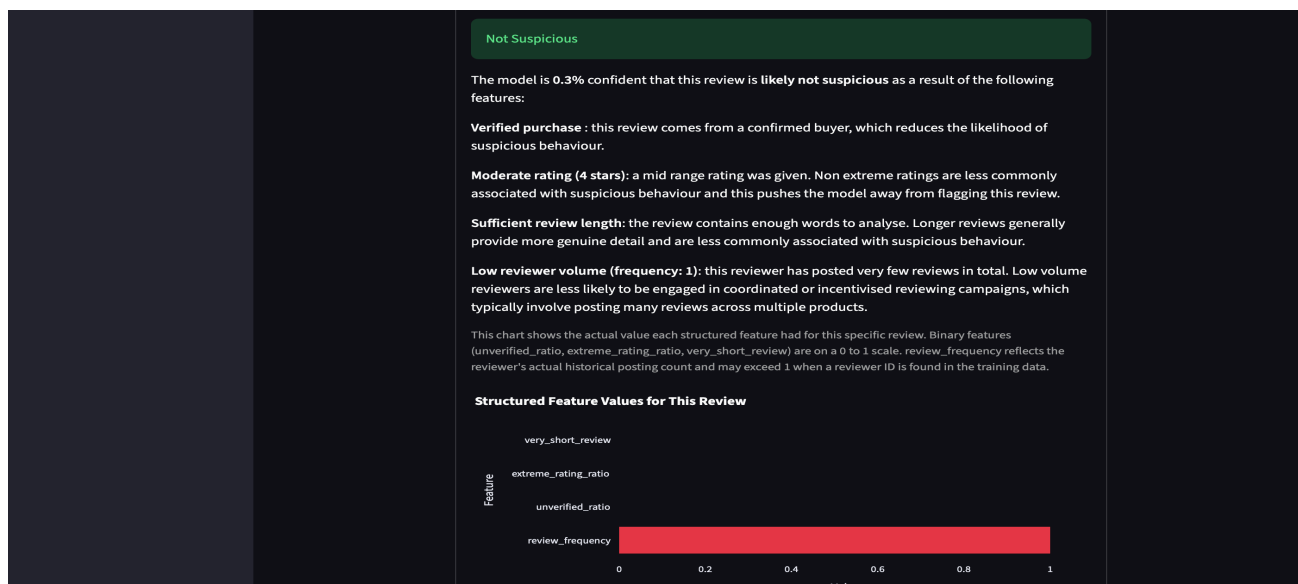
Limitations Identified

Despite the significant improvements in Stage 3, further limitations were identified. The word importance chart for the sentiment model included noise tokens, sub word fragments and punctuation produced by the DistilBERT tokeniser, which reduced the clarity and credibility of the word-level explanation. The SHAP interpretation logic also contained a directional inconsistency where values near zero were described with misleading direction labels. The interface also lacked a summary card presenting both model results immediately before the detailed breakdowns. These limitations were addressed in Stage 4.

Stage 4: Final Iteration

Model Retraining and SHAP Fix





Overview

Stage 4 addressed the most significant technical limitation identified during testing, the SHAP values for all four structured features showing as zero across every review analysed. This made the feature impact chart and plain English interpretations meaningless, undermining the application's core objective of explainable predictions. Stage 4 also introduced several UI improvements identified during real-world testing.

SHAP Zero Value Problem

Investigation revealed two root causes. First, the structured features were not scaled before being combined with TF-IDF features. Without StandardScaler, the structured features were numerically insignificant compared to the thousands of TF-IDF dimensions.

Second, the SHAP explainer was initialised with a single sample as the background distribution. When a single sample is used as the background, SHAP measures deviation from itself which is always zero.

The fix required applying StandardScaler to the structured features before combining with TF-IDF and using the full training data as the SHAP background distribution. A representative sample of 100 training observations was saved as shap_background.pkl and loaded at inference time to avoid loading the full dataset into the web application.

After these changes the structured features showed meaningful SHAP contributions: unverified_ratio at -0.4716, review_frequency at -0.2024 and extreme_rating_ratio at +0.2689 for the test review.

TF-IDF Vocabulary Filtering

Feature importance analysis revealed that the original TF-IDF vocabulary contained noisy words capturing product category patterns rather than genuine fraud signals. Words such as "downton", "outlander", "hannah" and "crate" appeared prominently despite having no relationship to suspicious reviewing behaviour. The vocabulary was filtered to retain only words with a meaningful frequency difference between suspicious and non-suspicious reviews, reducing the vocabulary from thousands of features to 278 discriminative words. This reduced bias and gave the structured features more competitive weight in the model.

UI Improvements

The following interface changes were made based on testing feedback. The summary card was added showing both model predictions immediately before the detailed breakdowns. Confidence gauge charts were placed side by side for compact comparison. Results were wrapped in expandable sections to reduce page density on first load. The Analyse and Clear buttons were placed side by side. A sidebar was added containing application description, model information, dataset statistics, team details and the reviewer ID field. A word count indicator was added below the text area. The model loading banner was replaced with a subtle caption. The reviewer frequency transparency note was updated to display only after the Analyse button is clicked rather than immediately on page load.

Feature Scale Caption Fix

The feature breakdown chart caption was updated to clarify that binary features operate on a 0 to 1 scale while review_frequency reflects the reviewer's actual historical posting count and may exceed 1 when a reviewer ID is found in the training data.

Noise Token Filtering

The word importance chart for the sentiment model was improved by filtering sub word tokens, punctuation, single characters and common English stop words using a regex filter that retains only pure alphabetic tokens of three or more characters. This removed noise tokens produced by the DistilBERT tokeniser that had no semantic value.

Key Features Added at This Stage

- StandardScaler applied to structured features before combining with TF-IDF
- SHAP background sample saved from training data and loaded at inference
- TF-IDF vocabulary filtered to 278 meaningful discriminative words
- Model retrained with scaling and filtered vocabulary
- Summary card showing both predictions before detailed breakdowns
- Side by side confidence gauge charts
- Expandable sections for sentiment and suspicious breakdowns
- Sidebar with application context, model information and reviewer ID
- Word count indicator
- Noise token filter for sentiment word importance chart
- Updated feature scale caption

Limitations Carried Forward

The SHAP chart labels are partially truncated on the chart due to margin constraints. The word importance chart occasionally surfaces incidental words that are not genuinely sentiment-bearing. The reviewer ID format mismatch between modern Amazon IDs and legacy training data IDs means the lookup table never fires for real-world reviews. The suspicious model confidence remains low due to the class imbalance in the training data.